Md. Khairul Basar

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Portfolio: https://k-basar.github.io/Portfolios/

EDUCATION

Bachelor of Science in Industrial and Production Engineering, (Feb 2020 – Aug 2025)
Shahjalal University of Science and Technology (SUST), Bangladesh
CGPA: 3.49 / 4.00

RESEARCH PROFILE

Research-oriented Industrial & Production Engineering graduate with a strong focus on Computational Biology, Machine Learning, and Data-Driven Optimization. Experienced in developing Predict-Then-Optimize frameworks and integrating ML models with mathematical optimization to solve complex resource-allocation and scheduling problems. Seeking an MSc in Computational Biology to apply expertise in algorithmic thinking and stochastic methods to genomic data, personalized medicine, and the UAE's biotechnology ecosystem.

RESEARCH INTERESTS

1. **AI and Machine Learning in Biomedicine:** Interested in applying deep learning and ensemble methods to biological datasets for pattern discovery and disease prediction. Focus on advancing data-driven intelligence to support automated clinical decision-making and adaptive healthcare systems.
2. **Predict-Then-Optimize Frameworks for Biological Systems:** Focus on integrating predictive ML models with mathematical optimization to create end-to-end decision-support pipelines for biotechnology. Interest in applying uncertainty-aware predictions to inform downstream optimization in drug discovery and resource-constrained healthcare environments.
3. **Algorithmic Optimization in Bioinformatics:** Interested in using stochastic gradient descent and optimization techniques to improve the computational efficiency of large-scale genomic models. Aim to use ML models to approximate complex cost functions and guide solvers in high-dimensional biological data environments.
4. **Data-Driven Healthcare Decision Support Systems:** Goal to design intelligent systems that transform raw clinical and operational data into actionable medical insights. Emphasis on using predictive modeling and visualization to assist clinicians and policymakers in dynamic health environments.
5. **Simulation of Complex Biological Processes:** Interest in simulation-based modeling to evaluate biological system performance under varying operational or environmental scenarios. Focus on combining discrete-event simulation with data-driven models to identify bottlenecks in healthcare infrastructure and test improvement strategies.

PUBLICATIONS

1. **Optimizing Worker–Task Allocation in a Fish Processing Factory Using a Predict-Then-Optimize Approach**
    - ARPN Journal of Engineering and Applied Sciences (doi: 10.59018/012617)
2. **A Bi-Objective Optimization Model for Real-World Task Allocation in RMG Sewing Lines**
    - 8th IEOM Bangladesh International Conference (**Best Track Paper,** doi: 10.46254/BA08.20250363)
3. **Data-Driven Prediction and Recommendation of Elective Courses for Master's Students**
    - 8th IEOM Bangladesh International Conference (doi: 10.46254/BA08.20250209s)
4. **Maintaining Lane Discipline Can Reduce the Waiting Time in a Road Intersection: A Simulation Study**
    - ICERIE 2025 (doi: https://doi.org/10.2991/978-94-6463-884-4_85)
5. **Determinants of CODP Positioning and Their Effects on Supply Chain Outcomes: An Empirical Investigation in the Furniture Sector**
    - ICERIE 2025 (doi: 10.46254/BA08.20250373)

Projects

1. Excel-ChatGPT Integration: Automating Web Scraping & AI Responses with VBA
    - Integrated ChatGPT with Excel VBA to automate web scraping and instant AI responses. Streamlined data extraction and enhanced workflow automation. Technologies: Excel VBA, OpenAI API, Web Scraping
2. Object Detection Safety System for Industrial Machinery
    - Designed an Object Detection System to enhance workplace safety. Prevents machinery-related hazards by identifying risky movements in real time. Technologies: Computer Vision, Arduino
3. Dual-Prediction System for Traffic Density & Vehicle Energy Consumption Using Machine Learning
    - Built a Python-based ML model for real-time traffic density and fuel consumption prediction. Aids in route optimization and fuel efficiency for urban transportation. Technologies: Python (Pandas, Scikit-learn), Machine Learning, Data Science
4. Pre-Assembly Line Analysssis and Line Balancing at Walton Hi-Tech Industries Ltd.
    - Conducted a detailed Time Study and Motion Analysis on the refrigerator pre-assembly line. Applied line balancing techniques to identify bottlenecks and redistribute workloads effectively. Achieved a significant improvement in line efficiency and reduced idle time. Technologies: Excel, Line Balancing, Work Study, Industrial Engineering
5. Bottleneck Analysis & Scheduling Optimization at Premium Fish & Agro Ind. Ltd.
    - Conducted a bottleneck analysis of the production floor using data-driven methods. Applied scheduling techniques to reduce delays and increase efficiency. Technologies: Excel, Operations Research, Scheduling Algorithms

INDUSTRIAL & JOB EXPERIENCE

Intern, Walton Hi-Tech Industries PLC (May 2025)

• Conducted time study, motion analysis, line balancing, Applied Lean tools

Remote Employee, Industrial 3D Solutions (Dec 2024 – Jun 2025)

• Developed Excel–VBA automation tools, Supported laboratory product supply coordination

Bangladesh Industrial Technical Assistance Centre (BITAC), Dhaka (12 days)

- Gained insights into conventional and non-conventional manufacturing processes.

Khadim Ceramics, Sylhet (1 day)

- Acquired exposure to industrial ceramic manufacturing processes.

TECHNICAL SKILLS

- **Programming & Data Science:** NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn, TensorFlow, PyTorch, LangChain, CrewAI
- **Machine Learning:** Statistical ML Methods, Deep Learning, NLP, Computer Vision, Graph Neural Networks (GNNs), Agentic Decision-Making, Agentic LLMs, AI Reasoning, RAG
- **Optimization & Simulation:** LP, NLP, MILP, Convex Optimization, Discrete Optimization, Simulation Modeling, Gurobi, Scheduling, Arena Simulation

CERTIFICATIONS

- Introduction to Data Science with Python – EDGE Bangladesh
- Introduction to Deep Learning with PyTorch – DataCamp
- Excel/VBA for Creative Problem Solving – University of Colorado Boulder
- Six Sigma White Belt – CSSC

Achievements

- Awarded a 60% scholarship for the **ISCEA Prize Case Competition 2024**
- Finalist in **IPE Case Quest 1.0**

Interests

Poetry, Chess, Reading, Travelling

References

- **Dr. Abul Mukid Mohammad Mukaddes**
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  - Department of Industrial & Production Engineering, SUST
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